

Discussion 8 Worksheet

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Formulas:

1. **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1}$
2. **Derivative of Sine:** $\frac{d}{dx}(\sin x) = \cos x$
3. **Derivative of Cosine:** $\frac{d}{dx}(\cos x) = -\sin x$
4. **Product Rule:** $\frac{d}{dx}(u \cdot v) = u'v + uv'$
5. **Quotient Rule:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$
6. **Chain Rule:** $\frac{d}{dx}(f(g(h(x)))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$
7. **Gradient of a Function:** $\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$
8. **Directional Derivative:** $D_{\mathbf{u}}f(x, y, z) = \nabla f(x, y, z) \cdot \mathbf{u} = |\nabla f(x, y, z)| |\mathbf{u}| \cos \theta$
9. **Second Order Partial Derivative Test:** Let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$.
 - If $D > 0$ and $f_{xx} > 0$, then (x_0, y_0) is a local minimum.
 - If $D > 0$ and $f_{xx} < 0$, then (x_0, y_0) is a local maximum.
 - If $D < 0$, then (x_0, y_0) is a saddle point.
 - If $D = 0$, the test is inconclusive.

Problem 1: Find the first and second order derivatives of the given function

$$f(x) = x^2 \cos(x) + \frac{2}{x}$$

Problem 2: Given $f(x) = ((x^2)^2)^2$. Find $f'(x)$ using chain rule.

Problem 3:

Given

$$f(x, y) = x^3 + y^3 - 3x^2 + 3y^2,$$

find all the critical points and determine whether they are saddle points, local maxima, or local minima using the second order partial derivative test.

Problem 4: The temperature T (in $^{\circ}\text{C}$) at a point (x, y, z) in space is given by the function:

$$T(x, y, z) = 4x^2y - 2y^2z + xyz$$

- Find the rate of change of temperature at the point $(1, -2, 3)$ in the direction of the vector $(2, 1, -1)$.
- In which direction does the temperature increase most rapidly at the point $(1, -2, 3)$?